Problem Set 2

Applied Stats II

Due: February 18, 2026

Instructions

- Please show your work! You may lose points by simply writing in the answer. If the problem requires you to execute commands in `R`, please include the code you used to get your answers. Please also include the `.R` file that contains your code. If you are not sure if work needs to be shown for a particular problem, please ask.
- Your homework should be submitted electronically on GitHub in `.pdf` form.
- This problem set is due before 23:59 on Wednesday February 18, 2026. No late assignments will be accepted.

We're interested in what types of international environmental agreements or policies people support (Bechtel and Scheve 2013). So, we asked 8,500 individuals whether they support a given policy, and for each participant, we vary the (1) number of countries that participate in the international agreement and (2) sanctions for not following the agreement.

Load in the data labeled `climateSupport.RData` on GitHub, which contains an observational study of 8,500 observations.

- Response variable:
  - `choice`: 1 if the individual agreed with the policy; 0 if the individual did not support the policy
- Explanatory variables:
  - `countries`: Number of participating countries [20 of 192; 80 of 192; 160 of 192]
  - `sanctions`: Sanctions for missing emission reduction targets [None, 5%, 15%, and 20% of the monthly household costs given 2% GDP growth]

Please answer the following questions:

1. Remember, we are interested in predicting the likelihood of an individual supporting a policy based on the number of countries participating and the possible sanctions for non-compliance.

   Fit an additive model. Provide the summary output, the global null hypothesis, and p-value. Please describe the results and provide a conclusion.

2. If any of the explanatory variables are statistically significant in this model, then:

   (a) For the policy in which nearly all countries participate [160 of 192], how does increasing sanctions from 5% to 15% change the odds that an individual will support the policy? (Interpretation of a coefficient)

   (b) For the policy in which very few countries participate [20 of 192], how does increasing sanctions from 5% to 15% change the odds that an individual will support the policy? (Interpretation of a coefficient)

   (c) What is the estimated probability that an individual will support a policy if there are 80 of 192 countries participating with no sanctions?

3. Would the answers to 2a and 2b potentially change if we included an interaction term in this model? Why?

   - Perform a test to see if including an interaction is appropriate.

Problem 1

Model and reference categories. I estimated a logistic regression of support for the policy on `countries` and `sanctions`, treating both explanatory variables as *unordered categorical* factors. The reference categories are `countries = ‘‘20 of 192’’` and `sanctions = ‘‘None’’`. Therefore, the intercept corresponds to the log-odds of support for the baseline scenario (20 of 192, None), and each reported coefficient is interpreted relative to that baseline holding the other variable constant.

Global null hypothesis test. The global null is that all slope coefficients are zero (i.e., neither `countries` nor `sanctions` is associated with support). Using a likelihood-ratio (deviance) test, I obtain $\chi^2(5) = 215.15$ with $p = 1.62 \times 10^{-44}$. I reject the global null; at least one category differs from the reference in support.

Regression output (additive model).

Table 1: Logistic Regression: Support for International Environmental Policy

	Dependent variable:
	choice
countries80 of 192	0.336***
	(0.054)
countries160 of 192	0.648***
	(0.054)
sanctions5%	0.192***
	(0.062)
sanctions15%	−0.133**
	(0.062)
sanctions20%	−0.304***
	(0.062)
Constant	−0.273***
	(0.054)
Observations	8,500
Log Likelihood	−5,784.130
Akaike Inf. Crit.	11,580.260
Note:	*p<0.1; **p<0.05; ***p<0.01

R code used:

```r
# Explore the data
head(climateSupport)
str(climateSupport)
summary(climateSupport)

# Convert countries and sanctions to unordered factors with specified baselines
# Baseline for countries: "20 of 192"
# Baseline for sanctions: "None"

climateSupport$countries <- factor(climateSupport$countries,
                                   levels = c("20 of 192", "80 of 192", "160 of 192"),
                                   ordered = FALSE)

climateSupport$sanctions <- factor(climateSupport$sanctions,
                                   levels = c("None", "5%", "15%", "20%"),
                                   ordered = FALSE)

# Verify the factor levels and baselines
levels(climateSupport$countries)
levels(climateSupport$sanctions)

# Fit an additive logistic regression model

model1 <- glm(choice ~ countries + sanctions,
              family = binomial(link = "logit"),
              data = climateSupport)

# Summary of the model
summary(model1)

# Extract and display the global null hypothesis test
# The null hypothesis: H0: all slopes = 0 (no effect of predictors)
# Test statistic: Null deviance - Residual deviance
null_deviance <- model1$null.deviance
residual_deviance <- model1$deviance
df_diff <- model1$df.null - model1$df.residual
test_stat <- null_deviance - residual_deviance
p_value <- pchisq(test_stat, df_diff, lower.tail = FALSE)

cat("Global Null Hypothesis Test (Likelihood Ratio Test):\n")
cat("H0: All slopes = 0 (no effect of predictors)\n")
cat("Test Statistic (Chi-squared):", test_stat, "\n")
cat("Degrees of Freedom:", df_diff, "\n")
cat("P-value:", formatC(p_value, format = "e", digits = 2), "\n")
```

Problem 2

Because there is *no interaction term*, the effect of changing `sanctions` is the same regardless of the value of `countries`. Formally, for two sanctions levels a and b:

$$\log\left(\frac{\text{odds}(\text{Supported} \mid a)}{\text{odds}(\text{Supported} \mid b)}\right) = \delta_a - \delta_b,$$

which does *not* depend on `countries`.

(a) **Countries = 160 of 192; sanctions 5% → 15%.** The change in log-odds is

$$\Delta = \delta_{15} - \delta_5 = -0.13325 - 0.19186 = -0.32510.$$

The corresponding odds ratio is $e^{\Delta} \approx 0.722$. Thus, increasing sanctions from 5% to 15% multiplies the odds of support by about 0.72 (a ≈ 27.8% decrease in odds), holding `countries` fixed at 160 of 192.

(b) **Countries = 20 of 192; sanctions 5% → 15%.** In an additive model, this odds ratio is identical to part (a):

$$OR = e^{\delta_{15}-\delta_5} \approx 0.722.$$

Interpretation: increasing sanctions from 5% to 15% decreases the odds of support by about 27.8%, holding `countries` fixed at 20 of 192.

(c) **Predicted probability for 80 of 192 countries and no sanctions.** For `countries = 80 of 192` and `sanctions = None` (reference), the linear predictor is:

$$\hat{\eta} = \hat{\beta}_0 + \hat{\beta}_{80} = -0.27266 + 0.33636 = 0.06370.$$

Applying the logistic transform gives:

$$\hat{p} = \frac{1}{1+e^{-\hat{\eta}}} \approx 0.5159.$$

Therefore, the estimated probability of support is approximately $\hat{p} \approx 0.516$ (51.6%).

R code used:

```r
# Extract coefficients for interpretation
coef_summary <- summary(model1)$coefficients
coef_summary

# Problem 2a: 160 countries, sanctions from 5% to 15%
sanctions_5pct_coef <- coef_summary["sanctions5%", "Estimate"]
sanctions_15pct_coef <- coef_summary["sanctions15%", "Estimate"]

# Change in log-odds when going from 5% to 15% is:
```

```r
log_odds_change_2a <- sanctions_15pct_coef - sanctions_5pct_coef
odds_ratio_2a <- exp(log_odds_change_2a)

cat("\n2a. Policy with 160 countries: Sanctions increase from 5% to 15%\n")
cat("Log-odds change:", log_odds_change_2a, "\n")
cat("Odds Ratio:", odds_ratio_2a, "\n")
cat("Percentage change in odds:", (odds_ratio_2a - 1) * 100, "%\n")

# Problem 2b: 20 countries, sanctions from 5% to 15%
# With an additive model (no interaction) and odds ratio being a non-conditional metric (i.e., doesn't depend on other predictors in the model), the effect of sanctions is the same regardless of the number of countries
# So the odds ratio is identical to 2a

cat("\n2b. Policy with 20 countries: Sanctions increase from 5% to 15%\n")
cat("Log-odds change:", log_odds_change_2a, "\n")
cat("Odds Ratio:", odds_ratio_2a, "\n")
cat("Percentage change in odds:", (odds_ratio_2a - 1) * 100, "%\n")
cat("NOTE: In an ADDITIVE model, this is identical to 2a.\n")

# Problem 2c: Predicted probability for 80 countries with no sanctions

# eta_hat = intercept + countries80 * 1 + sanctions_none * 0 (since None is reference)
intercept <- coef(model1)["(Intercept)"]
countries_80_coef <- coef(model1)["countries80 of 192"]
eta_hat <- intercept + countries_80_coef
eta_hat
# Apply logistic transformation
predicted_prob_2c <- 1 / (1 + exp(-eta_hat))

cat("\n2c. Predicted probability: 80 countries with no sanctions\n")
cat("Linear predictor (eta):", eta_hat, "\n")
cat("Predicted Probability:", predicted_prob_2c, "\n")
cat("Predicted Probability (percentage):", predicted_prob_2c * 100, "%\n")
```

Problem 3

(i) **Would the answers to 2(a) and 2(b) differ if we included an interaction?**
Yes, potentially. With a `countries` × `sanctions` interaction, the effect of increasing sanctions could vary across the `countries` categories, so the odds ratio for 5% → 15% could differ between "160 of 192" and "20 of 192".

(ii) **Test whether the interaction is appropriate**
I compared the additive model to a model including `countries:sanctions` using a likelihood-ratio test. The deviance improvement is 6.2928 on 6 degrees of freedom ($p = 0.3912$). I therefore do *not* find evidence that the interaction improves model fit; the additive model is adequate, which supports why 2(a) and 2(b) are the same in my results.

R code used:

```
# Fit model with interaction term
model_interaction <- glm(choice ~ countries + sanctions + countries:
    sanctions,
                          family = binomial(link = "logit"),
                          data = climateSupport)

# Test for interaction using likelihood ratio test
anova(model1, model_interaction, test = "Chisq")
```

Table 2: Model Comparison: Additive vs. Interaction Model

	Dependent variable:	
	choice	
	(1)	(2)
countries80 of 192	0.336***	0.376***
	(0.054)	(0.106)
countries160 of 192	0.648***	0.613***
	(0.054)	(0.108)
sanctions5%	0.192***	0.122
	(0.062)	(0.105)
sanctions15%	−0.133**	−0.097
	(0.062)	(0.108)
sanctions20%	−0.304***	−0.253**
	(0.062)	(0.108)
countries80 of 192:sanctions5%		0.095
		(0.152)
countries160 of 192:sanctions5%		0.130
		(0.151)
countries80 of 192:sanctions15%		−0.052
		(0.152)
countries160 of 192:sanctions15%		−0.052
		(0.153)
countries80 of 192:sanctions20%		−0.197
		(0.151)
countries160 of 192:sanctions20%		0.057
		(0.154)
Constant	−0.273***	−0.275***
	(0.054)	(0.075)
Observations	8,500	8,500
Log Likelihood	−5,784.130	−5,780.983
Akaike Inf. Crit.	11,580.260	11,585.970
Note:		*p<0.1; **p<0.05; ***p<0.01